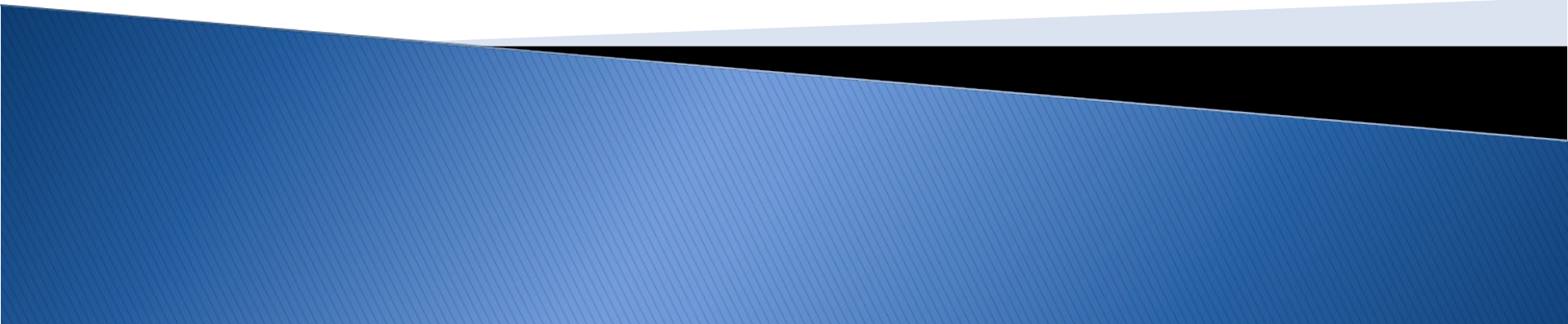


Introduction to Machine learning

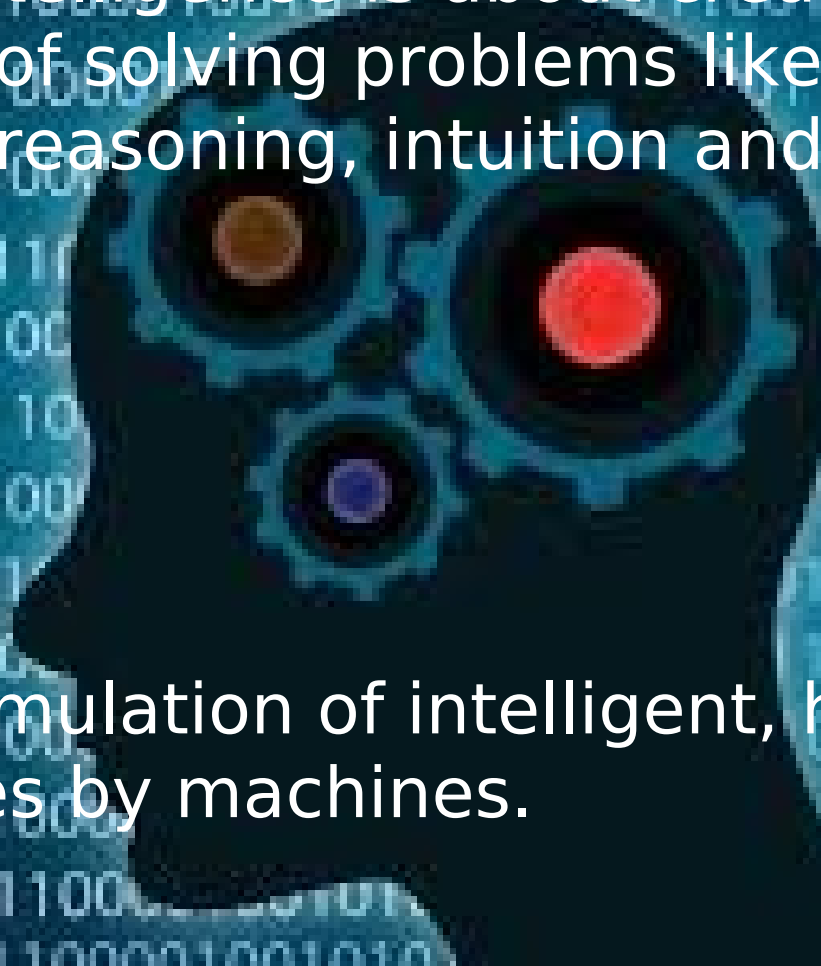
Escuela Politécnica Nacional
Facultad de Ingeniería de Sistemas
Profesor: Dr. Josafá Pontes



What is AI?

Artificial intelligence is about creating machines capable of solving problems like we do, through reasoning, intuition and creativity.

AI is the simulation of intelligent, human-like processes by machines.



Where is AI working?

Gaming

Natural
Language

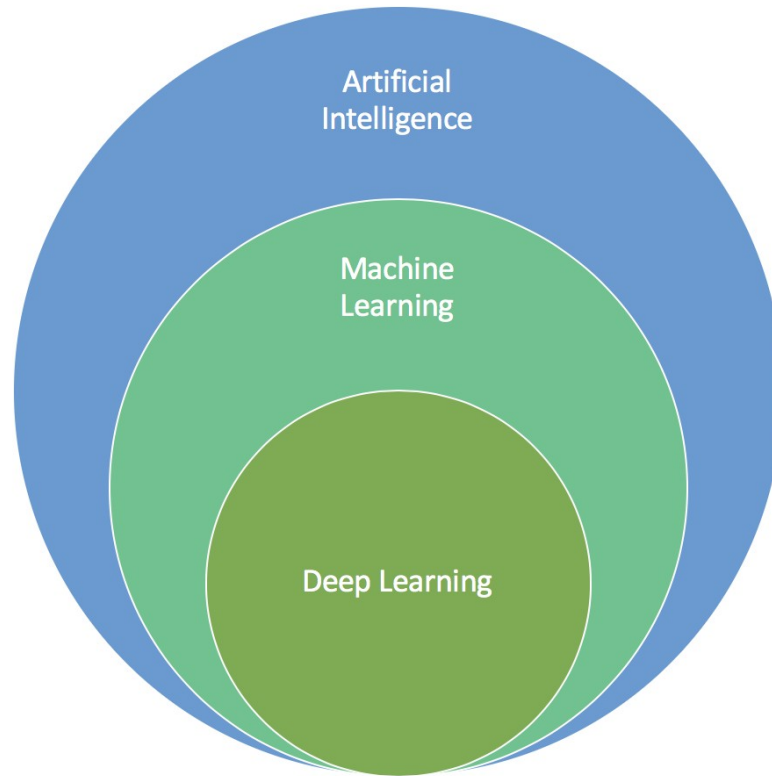
Machine
Learning

Robotics

Computer
Vision

Artificial Intelligence

Where is Machine Learning in AI?



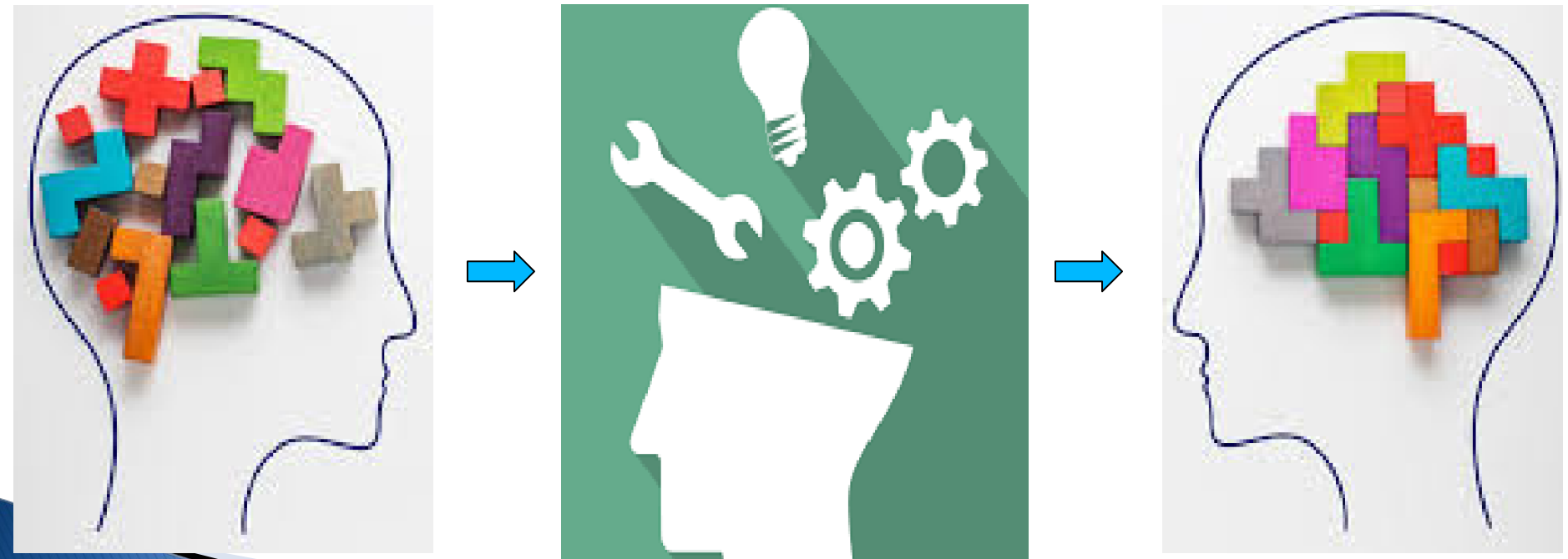
Is this Machine learning?



The learning process

“Learning is any process by which a system improves performance from experience.”

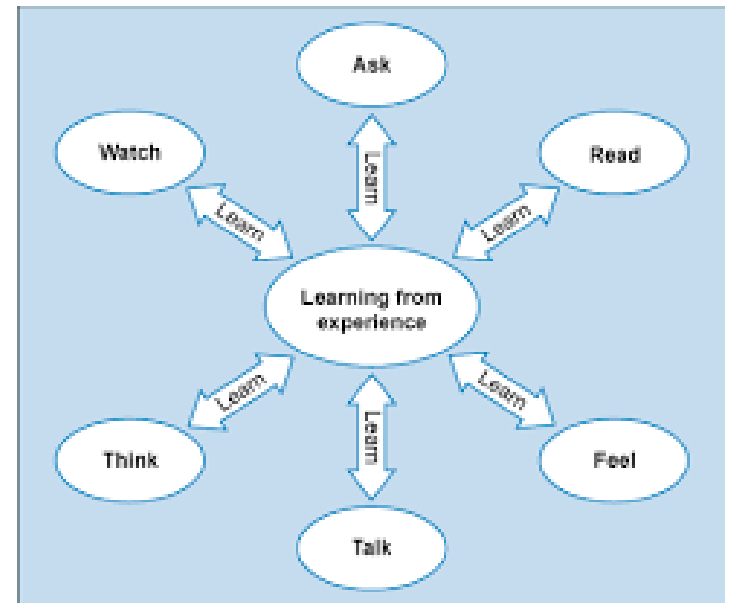
Herbert Simon (1978)



Learning as a human

“Learning is any process by which a system improves performance from experience.”

Herbert Simon (1978)



Learning as a machine

“Machine Learning is the study of algorithms that

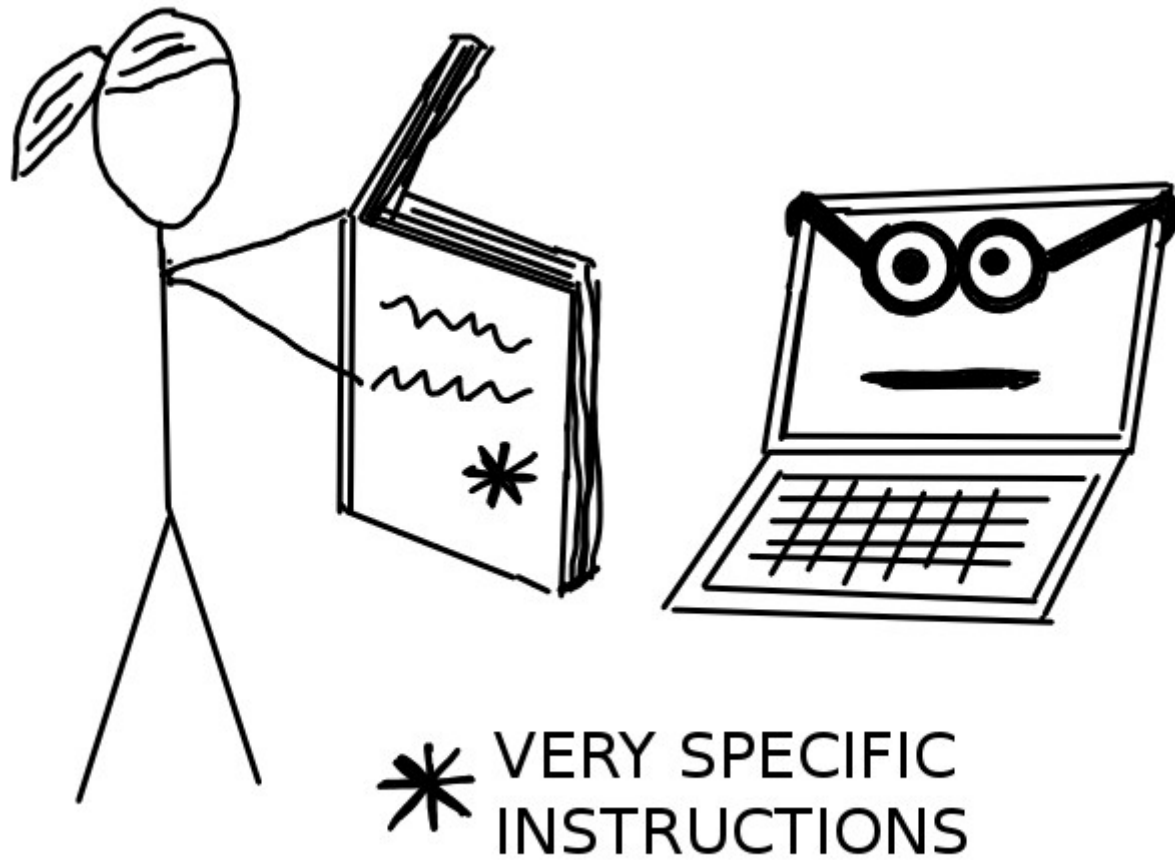
- Improve their performance P
- At some task T
- With experience E.

A well-defined learning task is given by

$\langle P, T, E \rangle$.”

Tom Mitchell (1998)

Is this Machine learning?



Traditional Programming

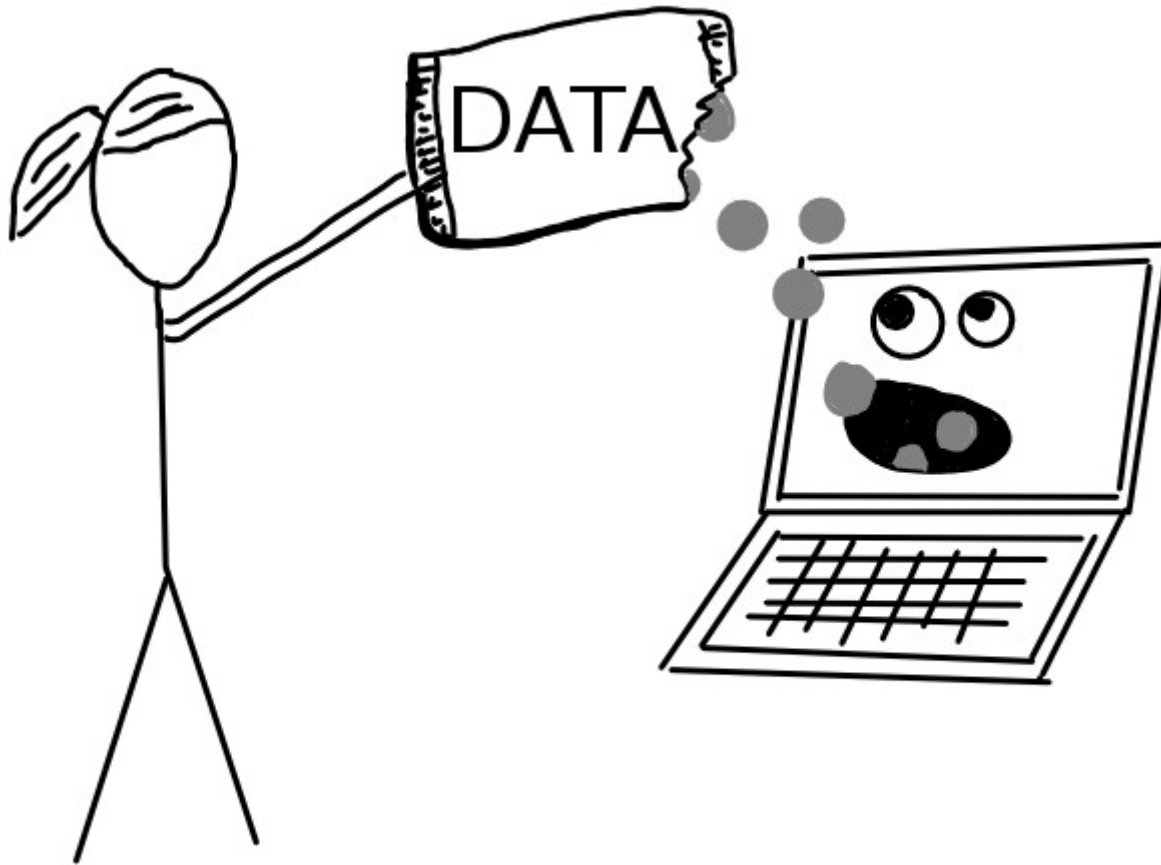


Learning as a machine

“Machine Learning is The Ability to machines to learn from the past experience, i.e. to learn from the data some correlation that help for taking a decision or doing predictions latter.”

Younes Charfaoui (2018)

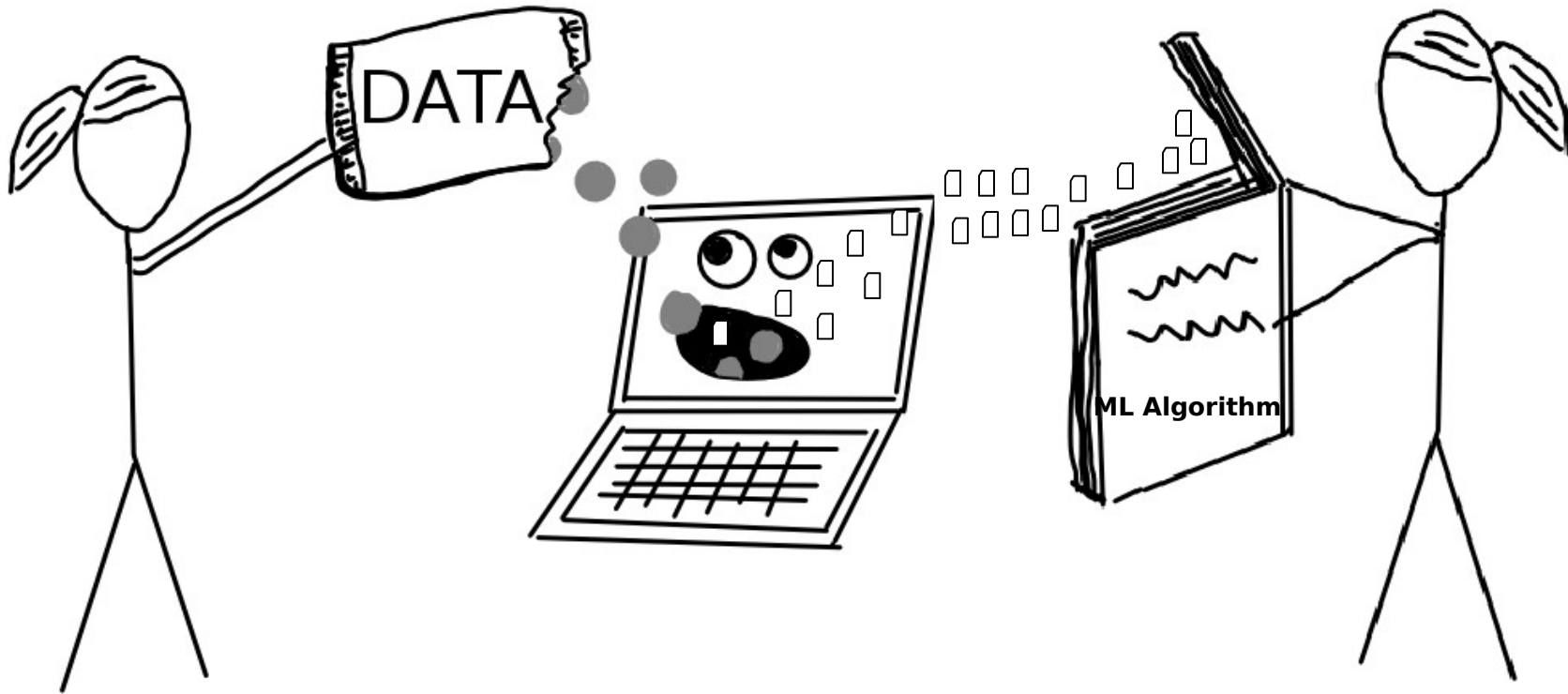
Is this Machine learning?



Machine Learning



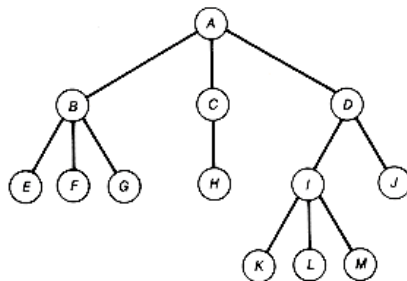
Is this Machine learning?



ML algorithms ingredients

$$\nabla f = \left[i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right] f$$

Calculus



Data structures

$$|\mathbf{A} - \lambda \cdot \mathbf{I}| = \begin{vmatrix} 0 & 1 \\ -2 & -3 \end{vmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = 0$$
$$\begin{vmatrix} -\lambda & 1 \\ -2 & -3 - \lambda \end{vmatrix} = \lambda^2 + 3\lambda + 2 = 0$$

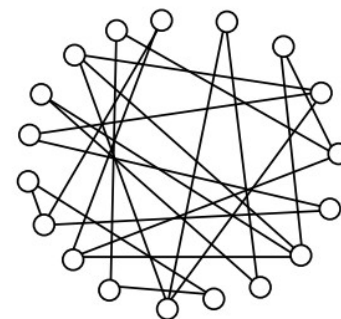
Linear algebra

$$\sum_{\substack{0 \leq i \leq m \\ 0 < j < n}} P(i, j)$$

Probability

$$\prod_{i=1}^n (1 - x_i) \geq 1 - \sum_{i=1}^n x_i$$

Statistics



Graph theory

ML tools



Caffe





What is Machine Learning ?

- Machine learning is an subfield of artificial intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed.
- Machine learning focuses on the development of computer programs that can access data and use it learn for themselves.



How Machine Learning Works?

- The Machine Learning process starts with inputting training data into the selected algorithm. Training data being known or unknown data to develop the final Machine Learning algorithm.
- To test whether this algorithm works correctly, new input data is fed into the Machine Learning algorithm. The prediction and results are then checked.



How Machine Learning Works?

- If the prediction is not as expected, the algorithm is re-trained multiple numbers of times until the desired output is found.
- This enables the Machine Learning algorithm to continually learn on its own and produce the most optimal answer that will gradually increase in accuracy over time.

Difference Between AI, ML, DL

ARTIFICIAL INTELLIGENCE

Artificial Intelligence (AI) is an umbrella discipline that covers everything related to making machines smarter.

MACHINE LEARNING

Machine Learning (ML) is commonly used along with AI but it is a subset of AI. ML refers to an AI system that can self-learn based on the algorithm.

DEEP LEARNING

Deep Learning (DL) is a machine learning applied to large data sets.

Types Of Machine Learning



Supervised Learning

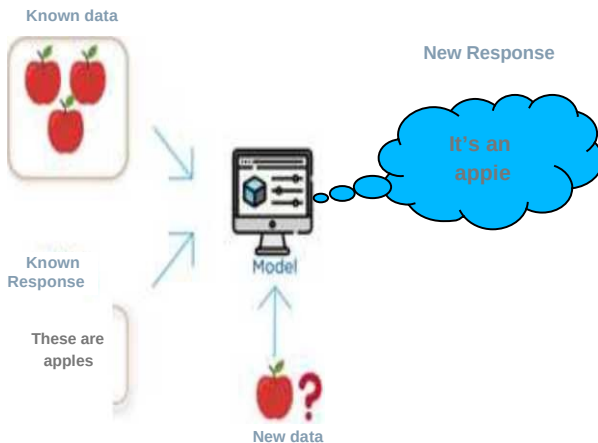


Unsupervised Learning



Reinforcement Learning

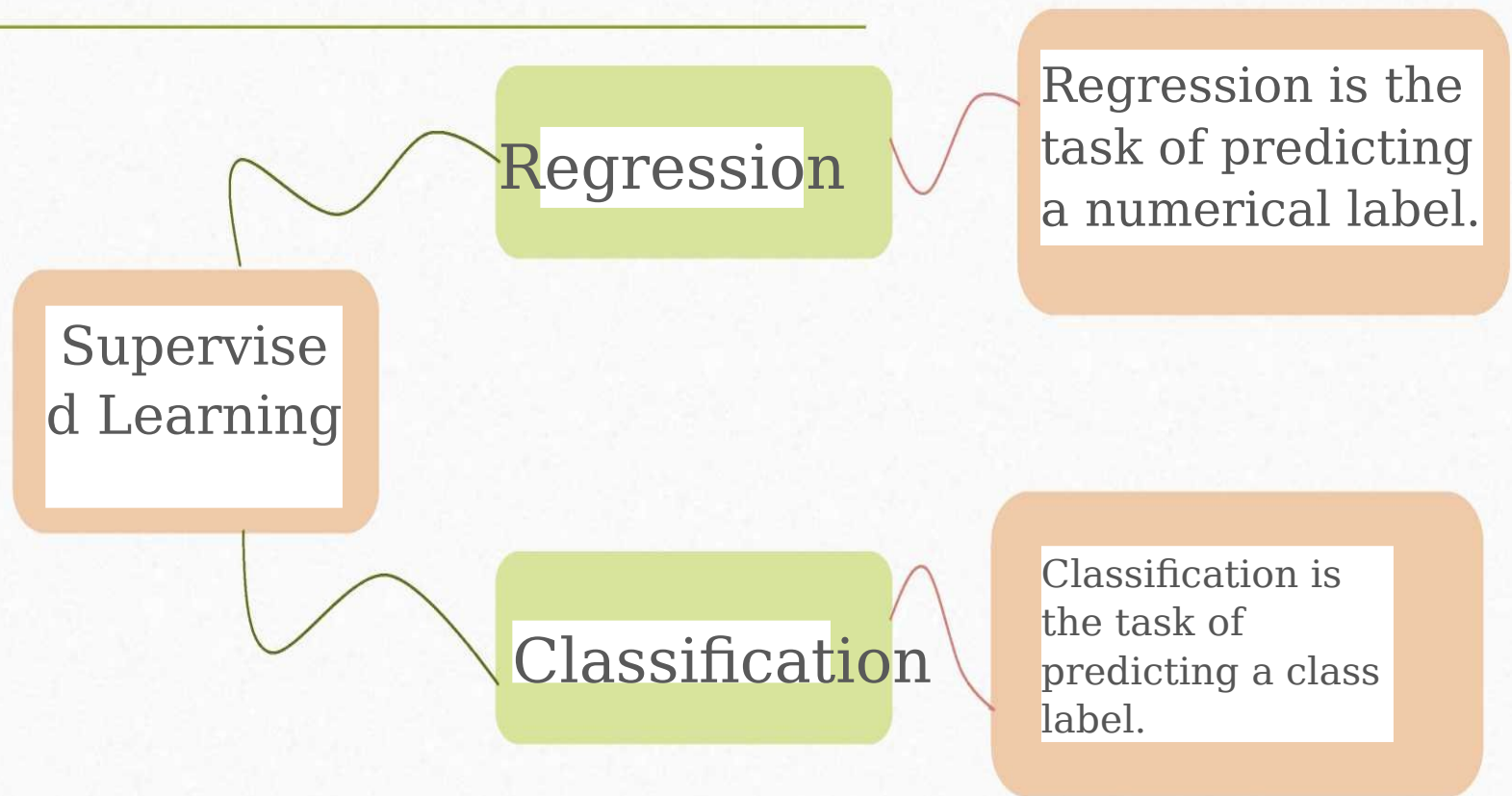
Supervised Learning



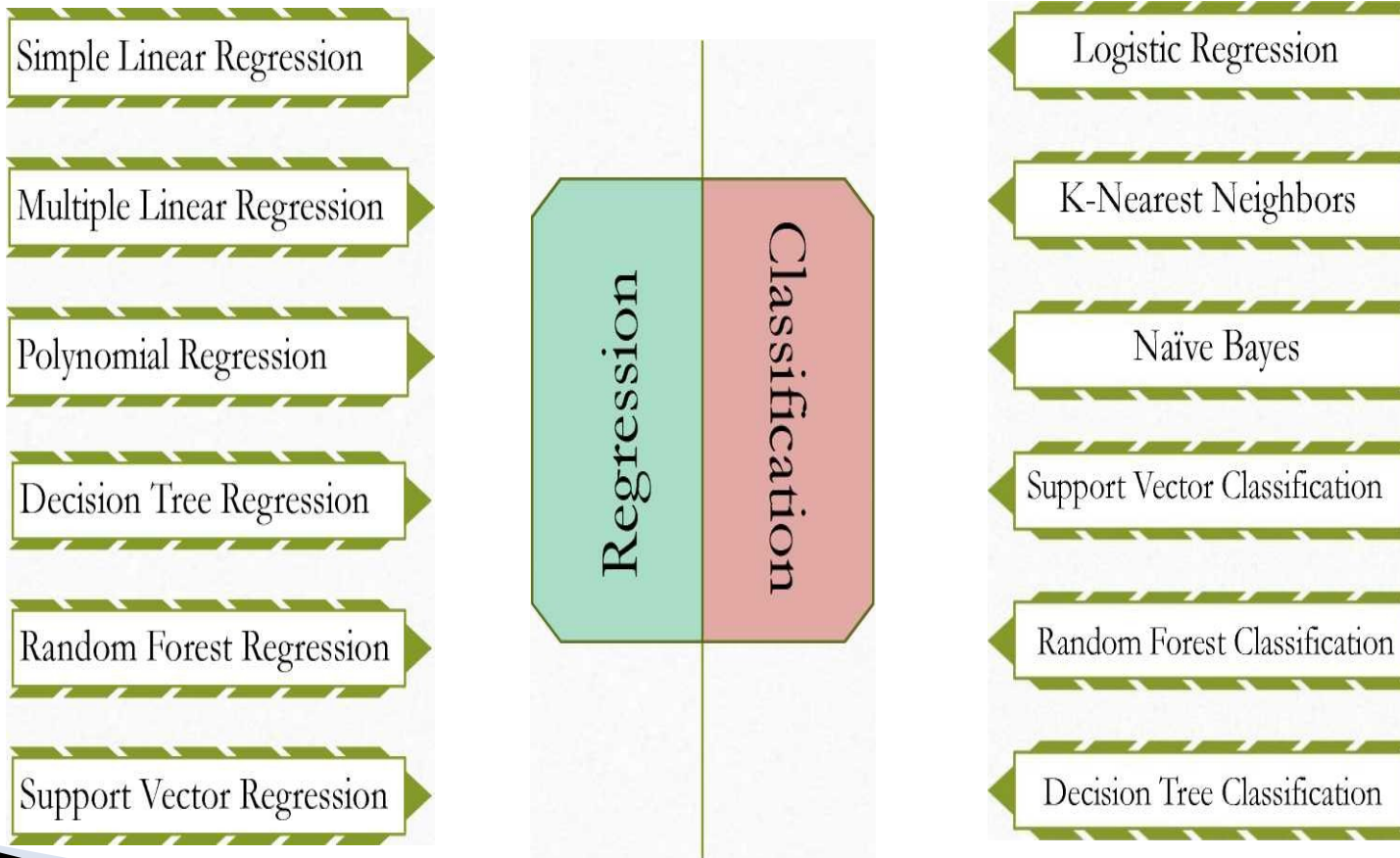
Supervised Learning describes a class of problem that involves using a model to learn a mapping between input examples and the target variable.

- Models are fit on training data comprised of inputs and outputs and used to make predictions on test sets where only the inputs are provided and the outputs from the model are compared to the withheld target variables and used to estimate the skill of the model.

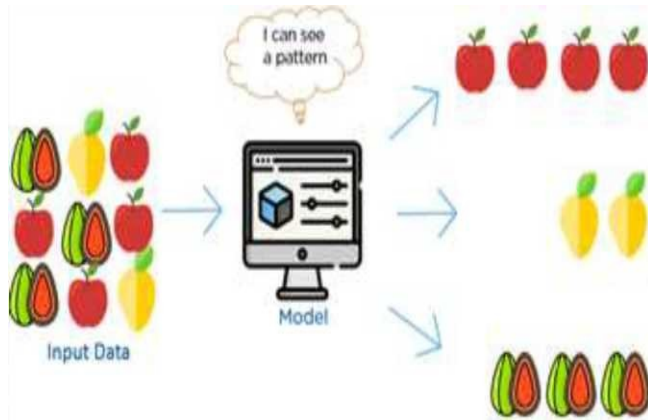
Supervised Learning problems can be divided into
two categories:



A few supervised Learning Algorithms



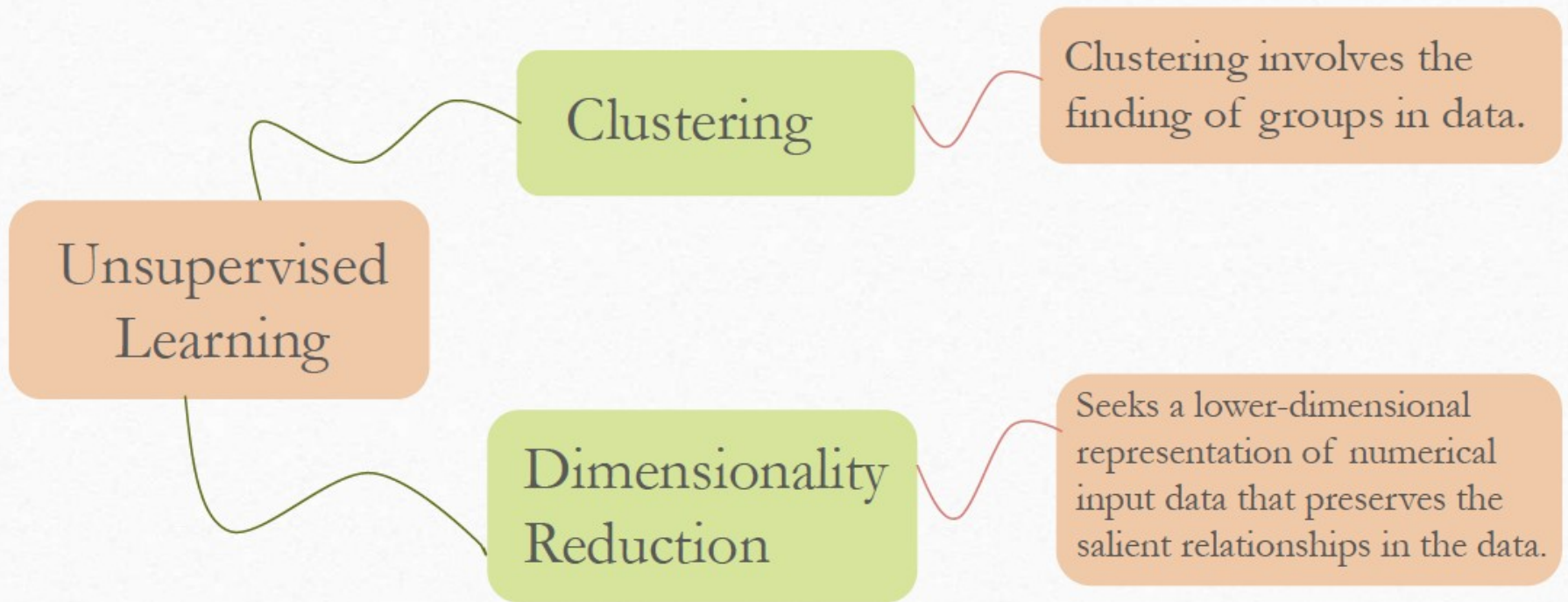
Unsupervised Learning



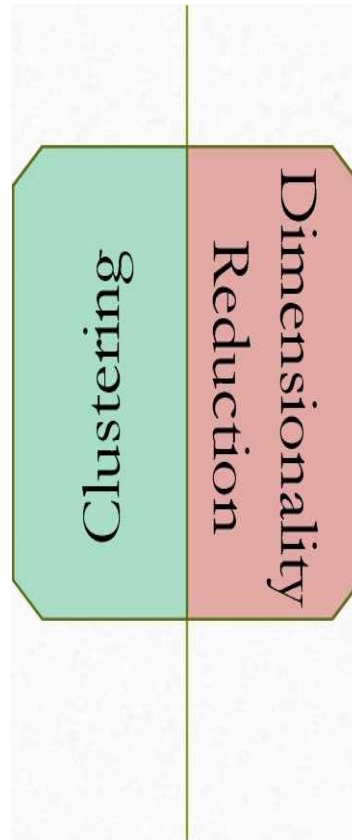
Unsupervised learning describes a class of problems that involves using a model to describe or extract relationships in data.

Compared to supervised learning, unsupervised learning operates upon only the input data without outputs or target variables. As such, unsupervised learning does not have a teacher correcting the model, as in the case of supervised learning.

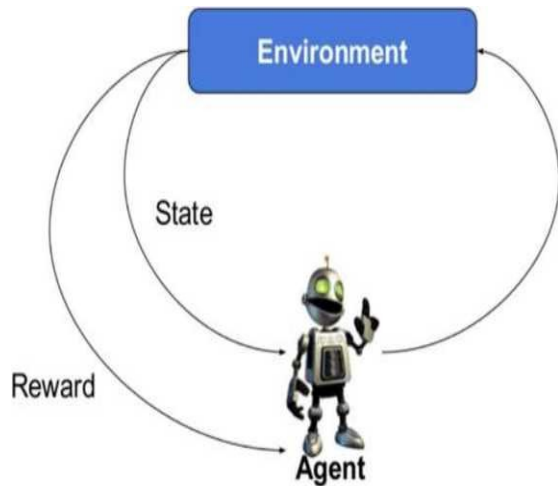
Unsupervised Learning problems can be divided
into two categories:



Unsupervised Learning



Reinforcement Learning



- Reinforcement learning describes a class of problems where an agent operates in an environment and must learn to operate using feedback.
- Reinforcement learning is learning what to do — how to map situations to actions—so as to maximize a numerical reward signal. The learner is not told which actions to take, but instead must discover which actions yield the most reward by trying them.

Applications Of Machine Learning

Supervised Learning

- Diagnostics
- Spam Detection
- Object-recognition
- Fraud Detection

Unsupervised Learning

- Recommender Systems
- Targeted Marketing
- Customer Segmentation
- Big Data Visualization

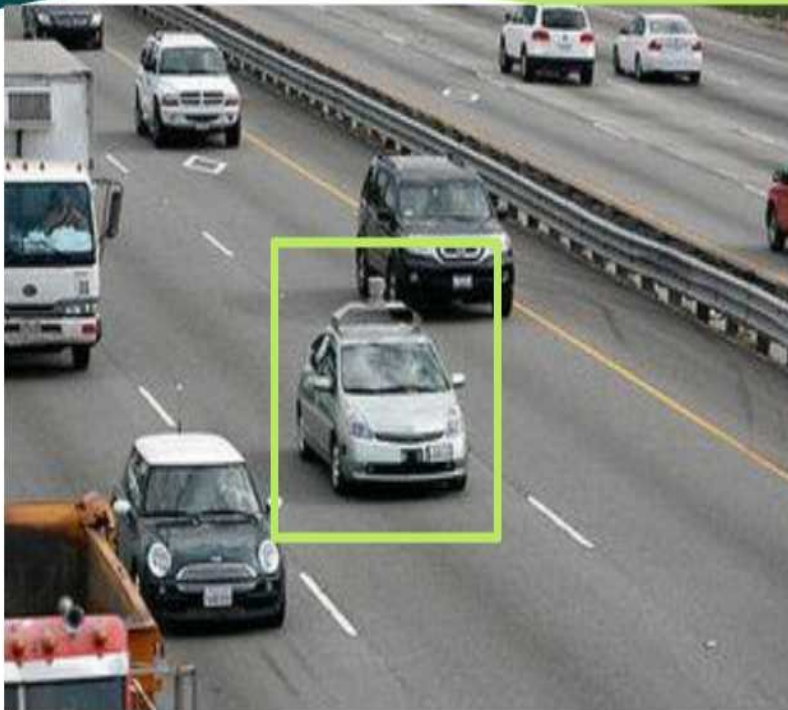
Reinforcement Learning

- Skill Acquisition
- Real-Time Decisions
- Robot Navigation
- Game AI

ML Applications



Autonomous Cars



We can teach a car to identify object on the road such as cars, persons, street, lane, etc..

Robotics

From self-driving cars to auto-navigating drones, AI & ML extends into many parts of robotics and transportation. The ultimate aim is to use location and sensor data to create vehicles capable of operating without human input.



Robotics

Always in
Agriculture, Robots
are trained to Crop
picking !!



Robotics

Robots are also playing great role in healthcare section !!



Image Recognition

The Amazing Dog face, ML to identify faces and then add in image of ears and noses in the correct places, that how money is made !!



Snapchat

Image Recognition

ML is used to track the neighboring object around self driving car

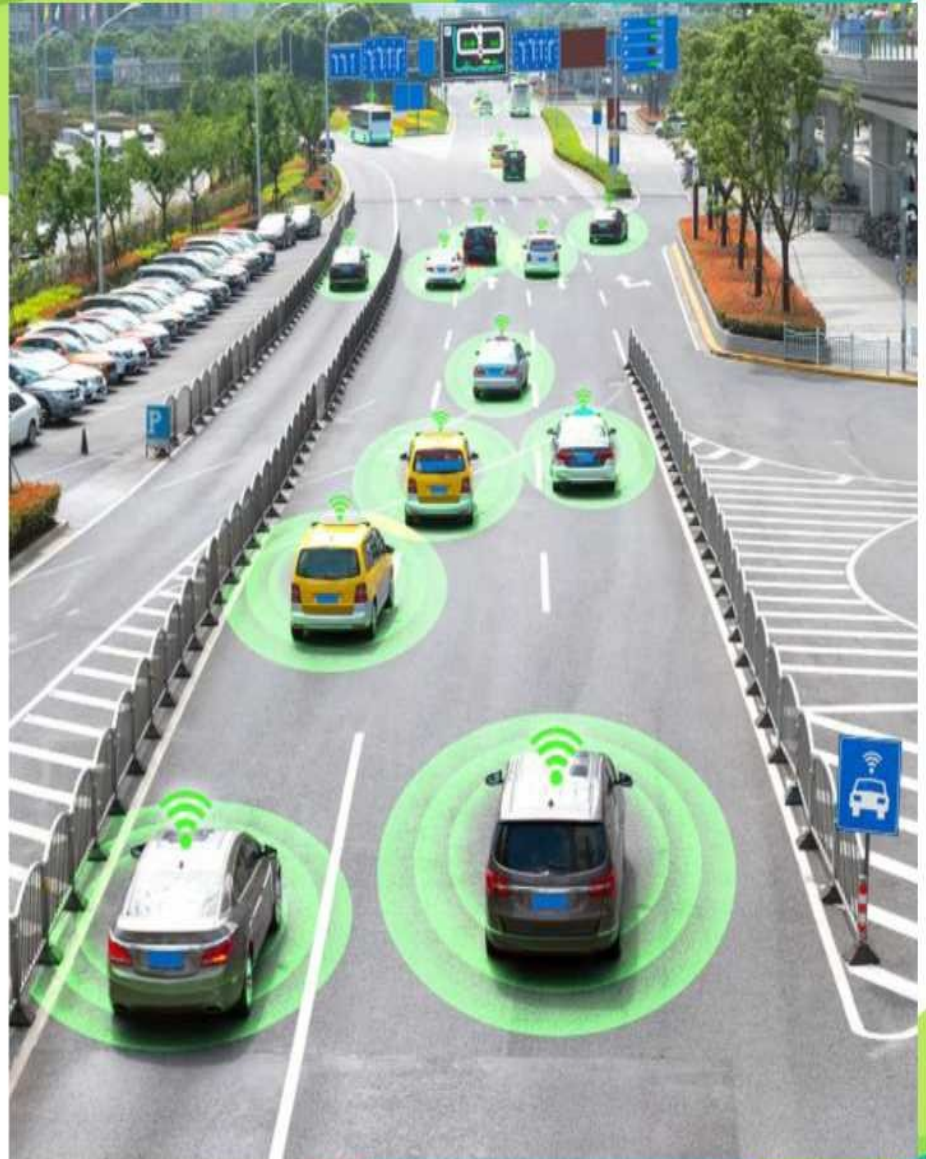


Image Recognition

ML is used in
Agriculture, Crop
Monitoring !!
Really amazing.



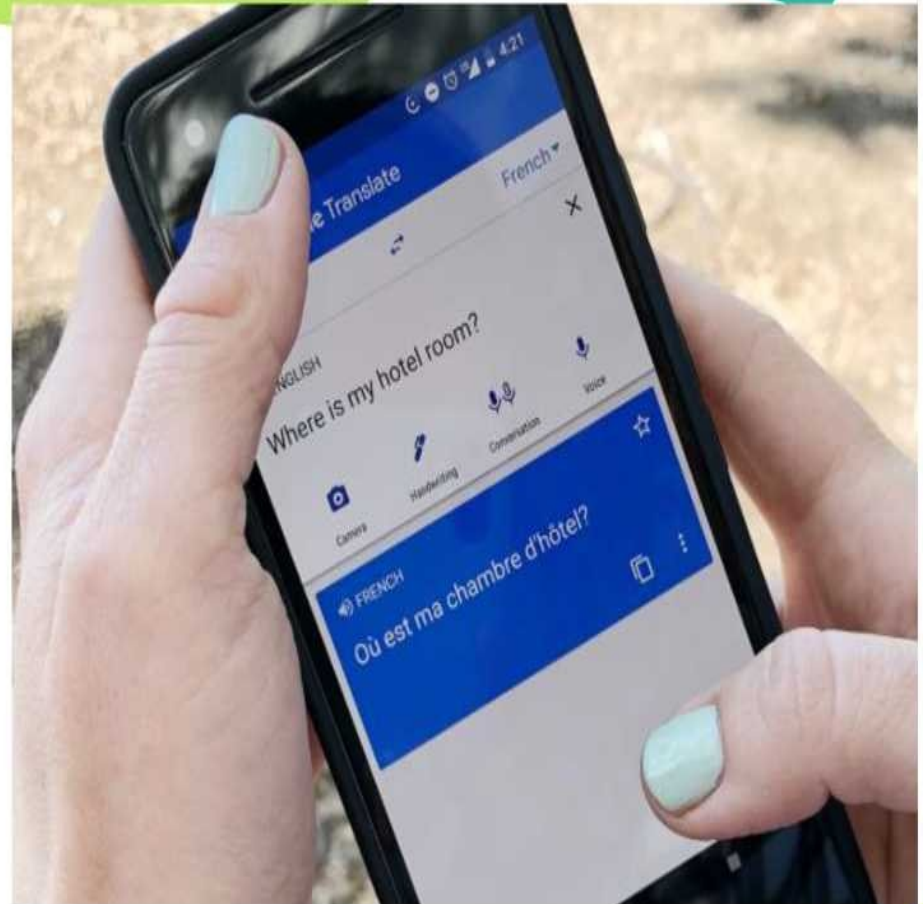
Natural Language Processing

The use of assistants and smart home can't be accomplished without NLP, Famous examples of NLP voice control include Apple's Siri, Amazon's Alexa, and Microsoft's Cortana.



Natural Language Processing

Translation is another side where ML & NLP played a great ball in it.



Natural Language Processing

Beside using NLP & ML into reading texts, some government are using in in Legal Document Processing.



Healthcare



Gogle's AI is better at predicting patient deaths than **hospitals**

Healthcare

With CV and ML,
diagnosis are
made easily and
accurately better
than expert
doctors.

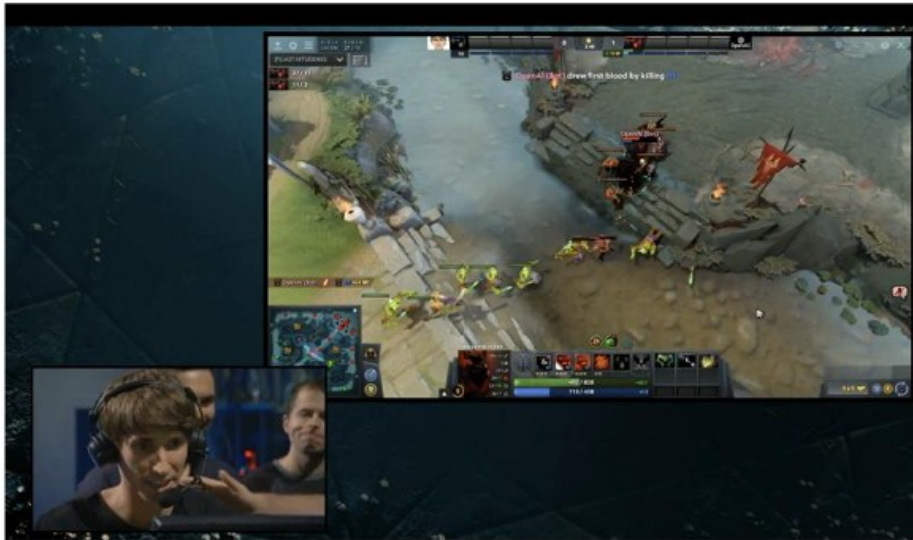


Gamins



Gaming

And finally The gaming section, where ML is also showing great results.



And Much More



Recommended literature

1 - Dive into Deep Learning. Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola. Release 0.16.3. <https://d2l.ai>

2 - Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition. Daniel Jurafsky. Stanford University. 3rd edition draft.

https://web.stanford.edu/~jurafsky/slp3/ed3book_dec302020.pdf

3 - Apache MXNET. Tutorials point. Simply easy learning.

https://www.tutorialspoint.com/apache_mxnet/apache_mxnet_tutorial.pdf

Recommended YouTube channels

1. Dot CSV
2. Statquest (Josh Starmer)
3. Siraj Raval
4. ritvikmath
5. Normalized Nerd
6. Zach Star

